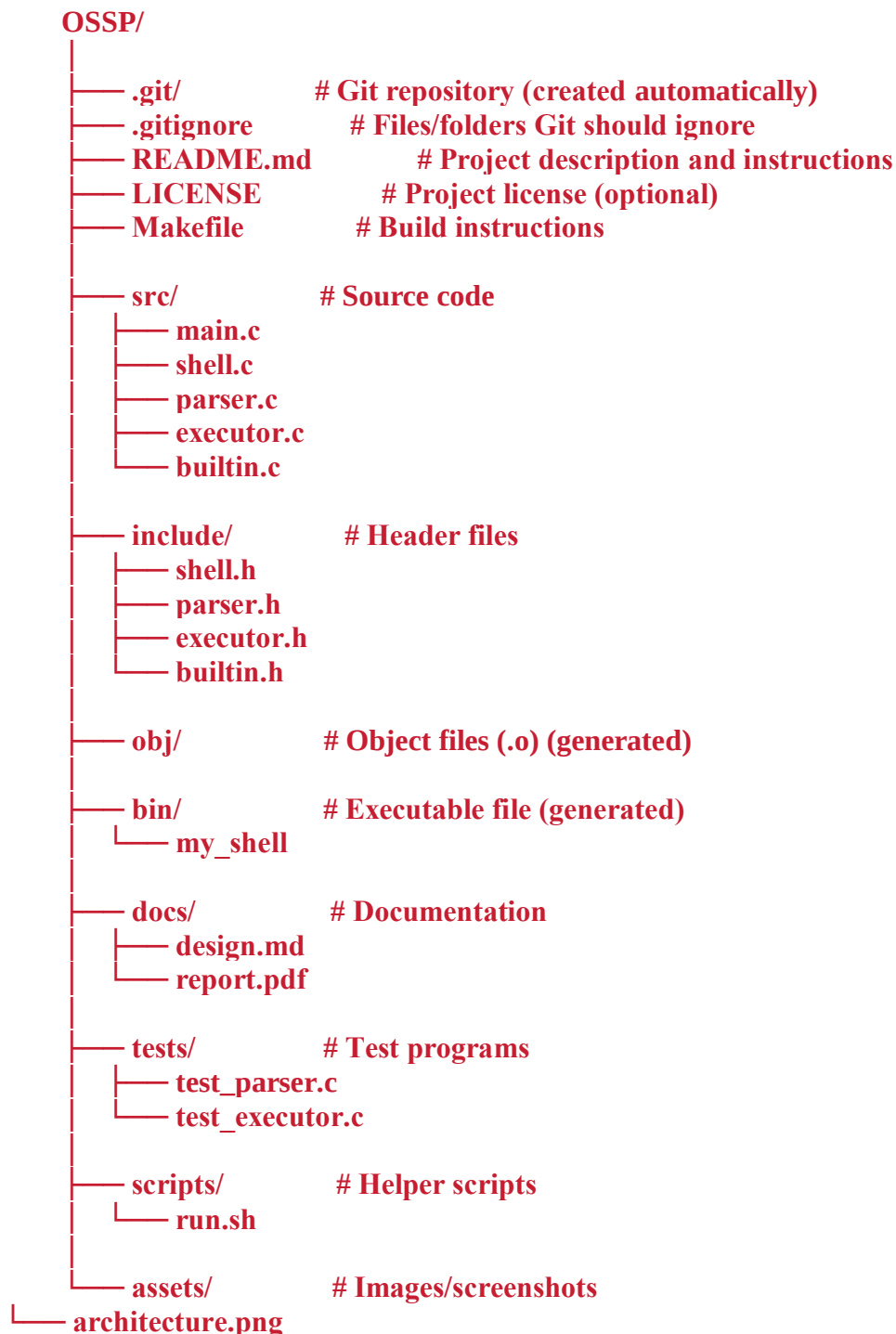


1. To Install Linux VM, Configure GCC, Setup Git Repository, Create Project Structure, Understand Shell Architecture, Build Initial Makefile.

Task -1: Create Project Structure as follows:



Note: For creating this structure use mkdir and touch commands.

```

divya@Divyasree:~$ pwd
/home/divya
divya@Divyasree:~$ mkdir OSSP
divya@Divyasree:~$ cd OSSP
divya@Divyasree:~/OSSP$ pwd
/home/divya/OSSP
divya@Divyasree:~/OSSP$ mkdir src include obj bin docs tests scripts assets
divya@Divyasree:~/OSSP$ ls
assets bin docs include obj scripts src tests
divya@Divyasree:~/OSSP$ touch .gitignore README.md LICENSE Makefile
divya@Divyasree:~/OSSP$ cd src
divya@Divyasree:~/OSSP/src$ touch main.c shell.c parser.c executor.c builtin.c
divya@Divyasree:~/OSSP/src$ cd ..
divya@Divyasree:~/OSSP$ cd include
divya@Divyasree:~/OSSP/include$ touch shell.h parser.h executor.h builtin.h
divya@Divyasree:~/OSSP/include$ cd ..
divya@Divyasree:~/OSSP$ cd docs
divya@Divyasree:~/OSSP/docs$ touch design.md report.pdf
divya@Divyasree:~/OSSP/docs$ cd ..
divya@Divyasree:~/OSSP$ cd tests
divya@Divyasree:~/OSSP/tests$ touch test_parser.c test_executor.c
divya@Divyasree:~/OSSP/tests$ cd ..
divya@Divyasree:~/OSSP$ cd scripts
divya@Divyasree:~/OSSP/scripts$ touch run.sh
divya@Divyasree:~/OSSP/scripts$ cd ..
divya@Divyasree:~/OSSP$ cd assets
divya@Divyasree:~/OSSP/assets$ touch architecture.png
divya@Divyasree:~/OSSP/assets$ cd ..\
>
> cd ..
-bash: cd: too many arguments
divya@Divyasree:~/OSSP/assets$ cd ..
divya@Divyasree:~/OSSP$ git init

divya@Divyasree:~/OSSP$ git init
hint: Using 'master' as the name for the initial branch. This default branch name
hint: is subject to change. To configure the initial branch name to use in all
hint: of your new repositories, which will suppress this warning, call:
hint:
hint:     git config --global init.defaultBranch <name>
hint:
hint: Names commonly chosen instead of 'master' are 'main', 'trunk' and
hint: 'development'. The just-created branch can be renamed via this command:
hint:
hint:     git branch -m <name>
Initialized empty Git repository in /home/divya/OSSP/.git/
divya@Divyasree:~/OSSP$ tree
Command 'tree' not found, but can be installed with:
sudo snap install tree # version 2.1.3+pkg-5852, or
sudo apt install tree # version 2.1.1-2ubuntu3.24.04.2
See 'snap info tree' for additional versions.
divya@Divyasree:~/OSSP$ sudo apt install tree
[sudo] password for divya:
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
The following NEW packages will be installed:
  tree
0 upgraded, 1 newly installed, 0 to remove and 70 not upgraded.
Need to get 47.4 kB of archives.
After this operation, 111 kB of additional disk space will be used.
Get:1 http://archive.ubuntu.com/ubuntu/noble-updates/universe amd64 tree amd64 2.1.1-2ubuntu3.24.04.2 [47.4 kB]
Fetched 47.4 kB in 2s (28.0 kB/s)
Selecting previously unselected package tree.
(Reading database ... 46635 files and directories currently installed.)
Preparing to unpack ../tree_2.1.1-2ubuntu3.24.04.2_amd64.deb ...
Unpacking tree (2.1.1-2ubuntu3.24.04.2) ...

```

```
Setting up tree (2.1.1-2ubuntu3.24.04.2) ...
Processing triggers for man-db (2.12.0-4build2) ...
divya@Divyasree:~/OSSP$ tree
```

```
.
├── LICENSE
├── Makefile
├── README.md
├── assets
│   └── architecture.png
├── bin
├── docs
│   ├── design.md
│   └── report.pdf
├── include
│   ├── builtin.h
│   ├── executor.h
│   ├── parser.h
│   └── shell.h
├── obj
├── scripts
│   └── run.sh
├── src
│   ├── builtin.c
│   ├── executor.c
│   ├── main.c
│   ├── parser.c
│   └── shell.c
└── tests
    ├── test_executor.c
    └── test_parser.c
```

9 directories, 18 files

```
divya@Divyasree:~/OSSP$ cd bin
```

```
divya@Divyasree:~/OSSP/bin$ touch my_shell
```

```
divya@Divyasree:~/OSSP/bin$ cd ..
```

```
divya@Divyasree:~/OSSP$ tree
```

```
.
├── LICENSE
├── Makefile
├── README.md
├── assets
│   └── architecture.png
├── bin
│   └── my_shell
├── docs
│   ├── design.md
│   └── report.pdf
├── include
│   ├── builtin.h
│   ├── executor.h
│   ├── parser.h
│   └── shell.h
├── obj
├── scripts
│   └── run.sh
├── src
│   ├── builtin.c
│   ├── executor.c
│   ├── main.c
│   ├── parser.c
│   └── shell.c
└── tests
    ├── test_executor.c
    └── test_parser.c
```

9 directories, 19 files

2. To Analyze process abstraction, execute fork(), understand exec() family, analyze parent-child relationships, inspect process tree, practice system call tracing.

Task-1: Using a manual command understand the fork() and exec() system calls

Task-2: Write a C program to create a new process and accepting the user commands using fork() and exec() system calls.

Welcome to Ubuntu 24.04.1 LTS (GNU/Linux 6.18.33.2-microsoft-standard-WSL2 x86_64)

* Documentation: <https://help.ubuntu.com>
* Management: <https://landscape.canonical.com>
* Support: <https://ubuntu.com/pro>

System information as of Mon Aug 3 03:33:28 UTC 2026

System load: 0.4 Processes: 66
Usage of /: 0.2% of 1006.85GB Users logged in: 0
Memory usage: 8% IPv4 address for eth0: 172.18.168.228
Swap usage: 0%

* Strictly confined Kubernetes makes edge and IoT secure. Learn how MicroK8s just raised the bar for easy, resilient and secure K8s cluster deployment.

<https://ubuntu.com/engage/secure-kubernetes-at-the-edge>

This message is shown once a day. To disable it please create the /home/divya/.hushlogin file.

```
divya@Divyasree:~$ ps
  PID TTY          TIME CMD
  362 pts/0    00:00:00 bash
  557 pts/0    00:00:00 ps
divya@Divyasree:~$ pstree
```

/home/divya/.hushlogin file.

```
divya@Divyasree:~$ ps
  PID TTY          TIME CMD
  362 pts/0    00:00:00 bash
  557 pts/0    00:00:00 ps
divya@Divyasree:~$ pstree
systemd--agetty
        |
        |--cron
        |--dbus-daemon
        |--init-systemd(Ub)
        |   |--SessionLeader--Relay(362)--bash+++
        |   |--init--{init}
        |   |--login--bash
        |   |--{init-systemd(Ub)}
        |--rsyslogd--3*[{rsyslogd}]
        |--systemd--(sd-pam)
        |--systemd-journal
        |--systemd-logind
        |--systemd-resolve
        |--systemd-timedat
        |--systemd-timesyn--{systemd-timesyn}
        |--systemd-udev
        |--unattended-upgr--{unattended-upgr}
        |--wsl-pro-service--7*[{wsl-pro-service}]
```

```
divya@Divyasree:~$ ps -aux
USER      PID %CPU %MEM    VSZ   RSS TTY      STAT START   TIME COMMAND
root         1  0.9  0.2 21696 13052 ?        Ss   03:33   0:01 /sbin/init
root         2  0.0  0.0   3180  2204 hvc0     SL+  03:33   0:00 /init
root         6  0.0  0.0   3196  2132 hvc0     SL+  03:33   0:00 plan9 --control-socket 7 --log-level 4 --server-fd 8 --pipe-f
root        52  0.3  0.2 50304 15856 ?        S<s  03:33   0:00 /usr/lib/systemd/systemd-journald
root       108  0.1  0.1  25416  6780 ?        Ss   03:33   0:00 /usr/lib/systemd/systemd-udev
systemd+   159  0.1  0.2  21344 13184 ?        Ss   03:33   0:00 /usr/lib/systemd/systemd-resolved
systemd+   160  0.0  0.1  91036  7968 ?        Ssl  03:33   0:00 /usr/lib/systemd/systemd-timesyncd
```

```
divya@Divyasree:~$ ps -aux
USER      PID %CPU %MEM    VSZ   RSS TTY      STAT START   TIME COMMAND
root         1  0.9  0.2 21696 13052 ?        Ss   03:33   0:01 /sbin/init
root         2  0.0  0.0   3180  2204 hvc0     SL+  03:33   0:00 /init
root         6  0.0  0.0   3196  2132 hvc0     SL+  03:33   0:00 plan9 --control-socket 7 --log-level 4 --server-fd 8 --pipe-f
root        52  0.3  0.2 50304 15856 ?        S<s  03:33   0:00 /usr/lib/systemd/systemd-journald
root       108  0.1  0.1  25416  6780 ?        Ss   03:33   0:00 /usr/lib/systemd/systemd-udev
systemd+   159  0.1  0.2  21344 13184 ?        Ss   03:33   0:00 /usr/lib/systemd/systemd-resolved
systemd+   160  0.0  0.1  91036  7968 ?        Ssl  03:33   0:00 /usr/lib/systemd/systemd-timesyncd
root       175  0.0  0.0   4244  2684 ?        Ss   03:33   0:00 /usr/sbin/cron -f -P
message+   176  0.0  0.0   9548  5016 ?        Ss   03:33   0:00 dbus-daemon --system --address=systemd: --nofork --nopidfile
root       192  0.0  0.1 17980  8852 ?        Ss   03:33   0:00 /usr/lib/systemd/systemd-logind
root       199  0.1  0.2 1755848 12868 ?       Ssl  03:33   0:00 /usr/libexec/wsl-pro-service -vv
root       234  0.0  0.0   3124  1972 tty1     Ss+  03:33   0:00 /sbin/agetty -o -p -- \u --noclear - linux
syslog     236  0.0  0.1 222516  6084 ?       Ssl  03:33   0:00 /usr/sbin/rsyslogd -n -iNONE
root       243  0.1  0.3 107032 23264 ?       Ssl  03:33   0:00 /usr/bin/python3 /usr/share/unattended-upgrades/unattended-up
root       360  0.0  0.0   3184  1108 ?        Ss   03:33   0:00 /init
root       361  0.0  0.0   3200  1248 ?        S   03:33   0:00 /init
divya      362  0.0  0.0   6076  5376 pts/0    Ss   03:33   0:00 -bash
root       363  0.0  0.0   6700  4656 pts/1    Ss   03:33   0:00 /bin/login -f
divya      409  0.1  0.1  20116 11316 ?        Ss   03:33   0:00 /usr/lib/systemd/systemd --user
divya      410  0.0  0.0   21160  3572 ?        S   03:33   0:00 (sd-pam)
divya      434  0.0  0.0   6080  5360 pts/1    S+   03:33   0:00 -bash
divya      574  0.0  0.0   9588  5064 pts/0    R+   03:35   0:00 ps -aux
divya@Divyasree:~$ ls
```

```
divya@Divyasree:~$ sudo apt install strace
[sudo] password for divya:
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
The following NEW packages will be installed:
  strace
0 upgraded, 1 newly installed, 0 to remove and 70 not upgraded.
Need to get 584 kB of archives.
After this operation, 2153 kB of additional disk space will be used.
Get:1 http://archive.ubuntu.com/ubuntu noble/main amd64 strace amd64 6.8-0ubuntu2 [584 kB]
Fetched 584 kB in 4s (160 kB/s)
Selecting previously unselected package strace.
(Reading database ... 46642 files and directories currently installed.)
Preparing to unpack .../strace_6.8-0ubuntu2_amd64.deb ...
Unpacking strace (6.8-0ubuntu2) ...
Setting up strace (6.8-0ubuntu2) ...
Processing triggers for man-db (2.12.0-4build2) ...
divya@Divyasree:~$ strace ls
execve("/usr/bin/ls", ["ls"], 0x7fffa4e13e80 /* 26 vars */) = 0
brk(NULL)                               = 0x6122359e7000
mmap(NULL, 8192, PROT_READ|PROT_WRITE, MAP_PRIVATE|MAP_ANONYMOUS, -1, 0) = 0x7724e7e3e000
access("/etc/ld.so.preload", R_OK)      = -1 ENOENT (No such file or directory)
openat(AT_FDCWD, "/etc/ld.so.cache", O_RDONLY|O_CLOEXEC) = 3
fstat(3, {st_mode=S_IFREG|0644, st_size=19087, ...}) = 0
mmap(NULL, 19087, PROT_READ, MAP_PRIVATE, 3, 0) = 0x7724e7e39000
close(3)                                 = 0
openat(AT_FDCWD, "/lib/x86_64-linux-gnu/libselinux.so.1", O_RDONLY|O_CLOEXEC) = 3
read(3, "\177ELF\2\1\1\0\0\0\0\0\0\0\0\0\3\0>\0\1\0\0\0\0\0\0\0\0\0\0\0\0...", 832) = 832
fstat(3, {st_mode=S_IFREG|0644, st_size=174472, ...}) = 0
```

```
close(1)                                = 0
close(2)                                = 0
exit_group(0)                           = ?
+++ exited with 0 +++
divya@Divyasree:~$ pid_t fork(void);
-bash: syntax error near unexpected token `('
divya@Divyasree:~$ nano process.c
divya@Divyasree:~$ gcc process.c -o process
divya@Divyasree:~$ ./process
```

```
GNU nano 7.2 process.c
#include <stdio.h>
#include <unistd.h>
#include <sys/types.h>
#include <sys/wait.h>
#include <stdlib.h>

int main()
{
    pid_t pid;
    char command[100];

    printf("Enter Linux command: ");
    scanf("%99s", command);

    pid = fork();

    if(pid < 0)
    {
        printf("Fork failed!\n");
        return 1;
    }
    else if(pid == 0)
    {
        printf("\nChild Process\n");
        printf("Child PID : %d\n", getpid());

        execlp(command, command, NULL);

        printf("Command not found!\n");
    }
}
```

^G Help ^O Write Out ^W Where Is ^K Cut ^T Execute ^C Location ^U Undo ^M Set Mark
 ^X Exit ^R Read File ^\ Replace ^U Paste ^J Justify ^_ Go To Line ^-E Redo ^-G Copy

```
divya@Divyasree:~$ nano process.c
divya@Divyasree:~$ gcc process.c -o process
divya@Divyasree:~$ ./process
Enter Linux command: ls
```

```
Parent Process
Parent PID : 789
Child PID  : 790
```

```
Child Process
Child PID : 790
OSSP  process  process.c
```

```
Child execution completed.
divya@Divyasree:~$ nano process.c
divya@Divyasree:~$ |
```

```
{
    printf("Fork failed!\n");
    return 1;
}
else if(pid == 0)
{
    printf("\nChild Process\n");
    printf("Child PID : %d\n", getpid());

    execlp(command, command, NULL);

    printf("Command not found!\n");
    exit(1);
}
else
{
    printf("\nParent Process\n");
    printf("Parent PID : %d\n", getpid());
    printf("Child PID : %d\n", pid);

    wait(NULL);

    printf("\nChild execution completed.\n");
}

return 0;
}
```